Supplementary Information 1: Detailed Analytical Proofs for "A Vorticity-Selective Dissipative Model ($\mathrm{NS}_{\mathrm{TZT}}$) for Incompressible Flows on the 3-Torus ($\mathbb{T}^3$): Global Strong Well-Posedness, Calderón–Zygmund Cubic Regularity, and Subgrid-Scale Laminar Shielding"

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Abstract

This supplementary document provides complete, self-contained mathematical derivations and step-by-step analytical proofs supporting Part 3 of the 2026 *Grand Unified Framework* series (*A Vorticity-Selective Dissipative Model ($\mathrm{NS}_{\mathrm{TZT}}$) for Incompressible Flows on the 3-Torus ($\mathbb{T}^3$)*). Specifically, we present: (i) the Fourier-analytic characterization of solenoidal Sobolev spaces $H^s_\sigma(\mathbb{T}^3)$, the periodic Biot–Savart operator, and the C^∞ Friedrichs mollifier construction of the vorticity-selective threshold switch $\Phi(|\boldsymbol{\omega}|/\Omega_c)$; (ii) componentwise proofs of pointwise orthogonality, exact kinetic energy neutrality $\langle F_{\text{axis}}(\boldsymbol{u},\boldsymbol{\omega}),\boldsymbol{u}\rangle_{L^2} = 0$, and global Lipschitz continuity ($L_{\beta,\varepsilon} = C_\Phi/\Omega_c + 2/\varepsilon$) of the regularized gyroscopic alignment operator; (iii) full spectral eigenvalue derivations for the Hessian tensor $DV(\boldsymbol{\omega}) = D^2\Psi(\boldsymbol{\omega})$ of the radially symmetric convex dissipation potential, establishing Minty–Browder monotonicity and the three-zone piecewise cubic coercivity lower bound $V(\boldsymbol{\omega})\cdot\boldsymbol{\omega} \geq C_O\ell^2(|\boldsymbol{\omega}|^3 - 8\Omega_c^3)$; (iv) index-by-index Levi-Civita derivation of the rotational-to-strain subgrid vector calculus identity $-\nabla\times(\nu_t\boldsymbol{\omega}) = \nabla\cdot(2\nu_t S(\boldsymbol{u})) - 2(\nabla\boldsymbol{u})^{\mathsf{T}}\nabla\nu_t$; and (v) expanded functional-analytic proofs of Grönwall-free $H^1_\sigma(\mathbb{T}^3)$ enstrophy closure, Aubin–Lions–Simon compactness in $L^{3/2}((0,T)\times\mathbb{T}^3)$, unconditional uniqueness, vanishing filter convergence ($\ell\to 0^+$) to Leray–Hopf weak solutions, and higher $H^m_\sigma(\mathbb{T}^3)$ bootstrap regularity. Each analytical proof is explicitly mapped to its corresponding mechanized formal verification module in `Supplementary_2_Formal_Verification_Lean4` and symbolic verification script in `Supplementary_3_CAS_Python_Proofs`.

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1 Solenoidal Sobolev Spaces on $\mathbb{T}^3$ and Smooth Switch Construction

In this section, we record the exact Fourier-analytic identities on the periodic three-dimensional torus $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3 \cong [0, 2\pi)^3$ and construct the C^∞ vorticity-selective switch $\Phi(\xi)$ corresponding to `Theorem0_Basic_Operators.lean` in `Supplementary_2`.

1.1 Fourier Characterization of $H^s_\sigma(\mathbb{T}^3)$ and Isometric Identities

Let $|\mathbb{T}^3| = (2\pi)^3$. Any vector field $\boldsymbol{u} \in L^2(\mathbb{T}^3; \mathbb{R}^3)$ is represented by its Fourier series

$$\boldsymbol{u}(x) = \sum_{k \in \mathbb{Z}^3} \hat{\boldsymbol{u}}_k e^{ik\cdot x}, \qquad \hat{\boldsymbol{u}}_k = \frac{1}{(2\pi)^3} \int_{\mathbb{T}^3} \boldsymbol{u}(x) e^{-ik\cdot x}\, \mathrm{d}x, \qquad \hat{\boldsymbol{u}}_{-k} = \overline{\hat{\boldsymbol{u}}_k}. \tag{1}$$

For $\boldsymbol{u} \in L^2_\sigma(\mathbb{T}^3)$, the zero-mean condition $\int_{\mathbb{T}^3} \boldsymbol{u}(x)\, \mathrm{d}x = \boldsymbol{0}$ implies $\hat{\boldsymbol{u}}_{\boldsymbol{0}} = \boldsymbol{0}$, and the distributional divergence-free condition $\nabla \cdot \boldsymbol{u} = 0$ is equivalent to the algebraic orthogonality $\boldsymbol{k} \cdot \hat{\boldsymbol{u}}_{\boldsymbol{k}} = 0$ for every $\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\boldsymbol{0}\}$.

Lemma 1.1 (Exact L^2 Gradient–Vorticity–Strain Isometric Identities on $\mathbb{T}^3$)**.** *For every solenoidal vector field $\boldsymbol{u} \in H^1_\sigma(\mathbb{T}^3)$ with vorticity $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ and symmetric rate-of-strain tensor $S(\boldsymbol{u}) = \frac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\mathsf{T}})$, the following exact $L^2(\mathbb{T}^3)$ norm identities hold:*

$$\|\nabla \boldsymbol{u}\|^2_{L^2(\mathbb{T}^3)} = \|\boldsymbol{\omega}\|^2_{L^2(\mathbb{T}^3)} = 2\|S(\boldsymbol{u})\|^2_{L^2(\mathbb{T}^3)} = (2\pi)^3 \sum_{\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\boldsymbol{0}\}} |\boldsymbol{k}|^2 |\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2. \tag{2}$$

Furthermore, for every $\boldsymbol{u} \in H^2_\sigma(\mathbb{T}^3)$, $-\Delta \boldsymbol{u} = \nabla \times \boldsymbol{\omega}$ pointwise, and $\|\Delta \boldsymbol{u}\|^2_{L^2(\mathbb{T}^3)} = \|\nabla \boldsymbol{\omega}\|^2_{L^2(\mathbb{T}^3)}$.

Proof. In Fourier space, $\widehat{(\nabla \times \boldsymbol{u})}_{\boldsymbol{k}} = i\boldsymbol{k} \times \hat{\boldsymbol{u}}_{\boldsymbol{k}}$. By Lagrange's vector identity in $\mathbb{C}^3$, for any $\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\boldsymbol{0}\}$:

$$|\boldsymbol{k} \times \hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2 = |\boldsymbol{k}|^2 |\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2 - |\boldsymbol{k} \cdot \hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2.$$

Since $\nabla \cdot \boldsymbol{u} = 0$ implies $\boldsymbol{k} \cdot \hat{\boldsymbol{u}}_{\boldsymbol{k}} = 0$, we have $|\widehat{\boldsymbol{\omega}}_{\boldsymbol{k}}|^2 = |\boldsymbol{k}|^2 |\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2 = \sum_{j=1}^3 |ik_j \hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2 = |\widehat{(\nabla \boldsymbol{u})}_{\boldsymbol{k}}|^2$. Summing over $\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\boldsymbol{0}\}$ via Plancherel's theorem establishes $\|\boldsymbol{\omega}\|^2_{L^2} = \|\nabla \boldsymbol{u}\|^2_{L^2}$. Next, expanding $2|S(\boldsymbol{u})|^2 = \frac{1}{2}(\partial_i u_j + \partial_j u_i)(\partial_i u_j + \partial_j u_i) = |\nabla \boldsymbol{u}|^2 + (\partial_i u_j)(\partial_j u_i)$ and integrating by parts twice on the boundaryless torus $\mathbb{T}^3$ ($\partial \mathbb{T}^3 = \emptyset$) gives:

$$\int_{\mathbb{T}^3} (\partial_i u_j)(\partial_j u_i)\, \mathrm{d}x = -\int_{\mathbb{T}^3} u_j \partial_j (\partial_i u_i)\, \mathrm{d}x = \int_{\mathbb{T}^3} (\nabla \cdot \boldsymbol{u})^2\, \mathrm{d}x = 0,$$

which proves $2\|S(\boldsymbol{u})\|^2_{L^2} = \|\nabla \boldsymbol{u}\|^2_{L^2}$. Applying the same argument to $\boldsymbol{\omega}$ (which satisfies $\nabla \cdot \boldsymbol{\omega} = \nabla \cdot (\nabla \times \boldsymbol{u}) = 0$) yields $\|\nabla \boldsymbol{\omega}\|^2_{L^2} = \|\nabla \times \boldsymbol{\omega}\|^2_{L^2} = \|\Delta \boldsymbol{u}\|^2_{L^2}$. □

1.2 Explicit Construction of the C^∞ Cutoff Switch $\Phi(\xi)$ and Solenoidality of F_{diss}

Lemma 1.2 (Existence and Derivative Bounds of the Smooth Cutoff Switch Φ)**.** *Define the standard non-negative bump kernel $\eta : \mathbb{R} \to [0, \infty)$ supported on $[1, 2]$ by*

$$\eta(r) := \begin{cases} \dfrac{1}{I_0} \exp\left(-\dfrac{1}{(r-1)(2-r)}\right), & \text{if } 1 < r < 2, \\ 0, & \text{if } r \le 1 \text{ or } r \ge 2, \end{cases} \qquad I_0 := \int_1^2 \exp\left(-\frac{1}{(r-1)(2-r)}\right) \mathrm{d}r > 0, \tag{3}$$

and define the cumulative integral function $\Phi(\xi) := \int_{-\infty}^{\xi} \eta(r)\, \mathrm{d}r$ for $\xi \in \mathbb{R}$. Then $\Phi \in C^\infty(\mathbb{R}; [0, 1])$ and satisfies:

1. *$\Phi(\xi) = 0$ identically for all $\xi \le 1$ (Laminar Shielding Zone);*

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2. $0 < \Phi(\xi) < 1$ *and* $\Phi'(\xi) = \eta(\xi) > 0$ *for all* $1 < \xi < 2$ *(Smooth Transition Shell);*

3. $\Phi(\xi) = 1$ *identically for all* $\xi \geq 2$ *(Supercritical Regularization Core);*

4. $0 \leq \Phi'(\xi) \leq C_\Phi := \eta(3/2) = I_0^{-1} e^{-4} < \infty$ *for all* $\xi \in \mathbb{R}$.

Proof. Because $(r-1)(2-r) > 0$ on $(1,2)$ and all derivatives of $\exp(-((r-1)(2-r))^{-1})$ vanish exponentially as $r \to 1^+$ and $r \to 2^-$, $\eta \in C_c^\infty(\mathbb{R})$ with $\operatorname{supp}(\eta) = [1,2]$ and $\int_{\mathbb{R}} \eta(r)\,\mathrm{d}r = 1$. By the fundamental theorem of calculus, $\Phi'(\xi) = \eta(\xi) \geq 0$, so $\Phi \in C^\infty(\mathbb{R})$ is non-decreasing, equals 0 for $\xi \leq 1$, and equals $\int_1^2 \eta(r)\,\mathrm{d}r = 1$ for $\xi \geq 2$. Since $(r-1)(2-r) = \frac{1}{4} - (r - \frac{3}{2})^2$ attains its unique maximum $\frac{1}{4}$ on $[1,2]$ at $r = \frac{3}{2}$, $\Phi'(\xi) = \eta(\xi)$ attains its global maximum $C_\Phi = \eta(3/2) = I_0^{-1} e^{-4}$ at $\xi = \frac{3}{2}$. □

Lemma 1.3 (Exact Distributional Solenoidality and Zero Mean of F_{diss})**.** *For any vorticity field* $\boldsymbol{\omega}$ *such that* $V(\boldsymbol{\omega}) = \nu_{\text{TZT}}(|\boldsymbol{\omega}|)\boldsymbol{\omega} \in L^1_{\text{loc}}(\mathbb{T}^3;\mathbb{R}^3)$*, the rotational dissipation operator* $F_{\text{diss}}(\boldsymbol{\omega}) = -\nabla \times V(\boldsymbol{\omega})$ *is divergence-free in* $\mathcal{D}'(\mathbb{T}^3)$ *and has zero spatial mean on* $\mathbb{T}^3$*. Consequently,* $\mathbb{P}F_{\text{diss}}(\boldsymbol{\omega}) = F_{\text{diss}}(\boldsymbol{\omega})$.

Proof. For any scalar test function $\phi \in C^\infty(\mathbb{T}^3)$, distributional pairing gives:

$$\begin{aligned}\langle \nabla \cdot F_{\text{diss}}(\boldsymbol{\omega}), \phi \rangle_{\mathcal{D}',\mathcal{D}} &= -\langle F_{\text{diss}}(\boldsymbol{\omega}), \nabla\phi \rangle = \int_{\mathbb{T}^3} \big(\nabla \times V(\boldsymbol{\omega})\big) \cdot \nabla\phi \,\mathrm{d}x \\ &= \int_{\mathbb{T}^3} V(\boldsymbol{\omega}) \cdot \big(\nabla \times (\nabla\phi)\big)\,\mathrm{d}x = 0,\end{aligned}$$

since $\nabla \times (\nabla\phi) \equiv \mathbf{0}$ by the symmetry of mixed second partial derivatives ($\varepsilon_{ijk}\partial_j\partial_k\phi = 0$). Taking the constant unit basis vector $\boldsymbol{e}_i$ ($i \in \{1,2,3\}$) on periodic $\mathbb{T}^3$ likewise yields:

$$\int_{\mathbb{T}^3} F_{\text{diss}}(\boldsymbol{\omega}) \cdot \boldsymbol{e}_i \,\mathrm{d}x = -\int_{\mathbb{T}^3} V(\boldsymbol{\omega}) \cdot (\nabla \times \boldsymbol{e}_i)\,\mathrm{d}x = 0.$$ □

2 Energy Neutrality and Lipschitz Analysis of the Gyroscopic Operator F_{axis}

This section provides the complete componentwise derivations for Lemma 3.1 and Lemma 3.2 of the main text, corresponding to `Theorem1_Gyroscopic_Neutrality.lean`, `Theorem5_Monotone_Uniqueness.lean`, and `CAS_Proof1_Gyroscopic_Neutrality.py`.

2.1 Componentwise Proof of Pointwise Orthogonality and $L^2(\mathbb{T}^3)$ Energy Neutrality

Proposition 2.1 (Pointwise and Global Energy Neutrality of F_{axis})**.** *Let* $\boldsymbol{u} \in L^2_\sigma(\mathbb{T}^3)$ *and let* $\boldsymbol{w} : \mathbb{T}^3 \to \mathbb{R}^3$ *be any measurable vector field. For any* $\varepsilon > 0$ *and* $\Omega_c > 0$*, define* $\hat{\boldsymbol{w}}_\varepsilon := (|\boldsymbol{w}|^2 + \varepsilon^2)^{-1/2}\boldsymbol{w}$ *and* $G(\boldsymbol{u},\boldsymbol{w}) := \Phi(|\boldsymbol{w}|/\Omega_c)(\boldsymbol{u} \times \hat{\boldsymbol{w}}_\varepsilon)$*. Then:*

1. *Pointwise on* $\mathbb{T}^3$, $G(\boldsymbol{u},\boldsymbol{w})(x) \cdot \boldsymbol{u}(x) = 0$ *and* $|G(\boldsymbol{u},\boldsymbol{w})(x)| \leq |\boldsymbol{u}(x)|$.

2. *Globally on* $L^2_\sigma(\mathbb{T}^3)$*, the Leray-projected gyroscopic force* $F_{\text{axis}}(\boldsymbol{u},\boldsymbol{w}) = c_X\Omega_c\mathbb{P}G(\boldsymbol{u},\boldsymbol{w})$ *satisfies:*

$$\langle F_{\text{axis}}(\boldsymbol{u},\boldsymbol{w}), \boldsymbol{u} \rangle_{L^2(\mathbb{T}^3)} = 0, \qquad \|F_{\text{axis}}(\boldsymbol{u},\boldsymbol{w})\|_{L^2(\mathbb{T}^3)} \leq |c_X|\Omega_c\|\boldsymbol{u}\|_{L^2(\mathbb{T}^3)}. \tag{4}$$

Proof. Let $\boldsymbol{u} = (u_1,u_2,u_3)^\mathsf{T}$ and $\hat{\boldsymbol{w}}_\varepsilon = (\hat{w}_1,\hat{w}_2,\hat{w}_3)^\mathsf{T}$. Expanding the Euclidean cross product componentwise yields:

$$\boldsymbol{u} \times \hat{\boldsymbol{w}}_\varepsilon = (u_2\hat{w}_3 - u_3\hat{w}_2,\ u_3\hat{w}_1 - u_1\hat{w}_3,\ u_1\hat{w}_2 - u_2\hat{w}_1)^\mathsf{T}.$$

Taking the Euclidean inner product with $\boldsymbol{u}$ gives:

$$(\boldsymbol{u} \times \hat{\boldsymbol{w}}_\varepsilon) \cdot \boldsymbol{u} = u_1u_2\hat{w}_3 - u_1u_3\hat{w}_2 + u_2u_3\hat{w}_1 - u_2u_1\hat{w}_3 + u_3u_1\hat{w}_2 - u_3u_2\hat{w}_1 \equiv 0.$$

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Multiplying by the scalar factor $\Phi(|\boldsymbol{w}(x)|/\Omega_c)$ gives $G(\boldsymbol{u},\boldsymbol{w})(x)\cdot\boldsymbol{u}(x)=0$ identically at every point $x\in\mathbb{T}^3$ (verified symbolically in `CAS_Proof1_Gyroscopic_Neutrality.py` and formally in `Theorem1_Gyroscopic_Neutrality.lean`).

Next, since $\varepsilon>0$, we have $|\hat{\boldsymbol{w}}_\varepsilon|^2=\frac{|\boldsymbol{w}|^2}{|\boldsymbol{w}|^2+\varepsilon^2}<1$, and by Lemma 1.2, $0\le\Phi(|\boldsymbol{w}|/\Omega_c)\le 1$. By Lagrange's identity $|\boldsymbol{u}\times\hat{\boldsymbol{w}}_\varepsilon|^2=|\boldsymbol{u}|^2|\hat{\boldsymbol{w}}_\varepsilon|^2-(\boldsymbol{u}\cdot\hat{\boldsymbol{w}}_\varepsilon)^2\le|\boldsymbol{u}|^2|\hat{\boldsymbol{w}}_\varepsilon|^2<|\boldsymbol{u}|^2$, it follows that $|G(\boldsymbol{u},\boldsymbol{w})(x)|\le|\boldsymbol{u}(x)|$ pointwise.

Finally, because $\mathbb{P}$ is an orthogonal projection onto $L^2_\sigma(\mathbb{T}^3)$ ($\mathbb{P}^*=\mathbb{P}$, $\|\mathbb{P}\|_{\mathrm{op}}=1$) and $\boldsymbol{u}\in L^2_\sigma(\mathbb{T}^3)$ satisfies $\mathbb{P}\boldsymbol{u}=\boldsymbol{u}$, we obtain:

$$\begin{aligned}\langle F_{\mathrm{axis}}(\boldsymbol{u},\boldsymbol{w}),\boldsymbol{u}\rangle &= c_X\Omega_c\langle\mathbb{P}G(\boldsymbol{u},\boldsymbol{w}),\boldsymbol{u}\rangle = c_X\Omega_c\langle G(\boldsymbol{u},\boldsymbol{w}),\mathbb{P}\boldsymbol{u}\rangle\\ &= c_X\Omega_c\int_{\mathbb{T}^3}G(\boldsymbol{u},\boldsymbol{w})(x)\cdot\boldsymbol{u}(x)\,\mathrm{d}x=0,\end{aligned}$$

and $\|F_{\mathrm{axis}}(\boldsymbol{u},\boldsymbol{w})\|_{L^2}\le|c_X|\Omega_c\|\mathbb{P}\|_{\mathrm{op}}\|G(\boldsymbol{u},\boldsymbol{w})\|_{L^2}\le|c_X|\Omega_c\|\boldsymbol{u}\|_{L^2}$. □

2.2 Spectral Eigenvalue Analysis of $D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon$ and Global Lipschitz Constant $L_{\beta,\varepsilon}$

Proposition 2.2 (Exact Jacobian Eigenvalues and Lipschitz Bound of the Regularized Axis Map)**.** *For any fixed $\varepsilon>0$, the regularized unit direction map $\boldsymbol{w}\mapsto\hat{\boldsymbol{w}}_\varepsilon:=(|\boldsymbol{w}|^2+\varepsilon^2)^{-1/2}\boldsymbol{w}$ belongs to $C^\infty(\mathbb{R}^3;\mathbb{R}^3)$. Its symmetric Jacobian matrix $D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon(\boldsymbol{w})$ is given by*

$$(D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon(\boldsymbol{w}))_{ij}=\frac{\partial(\hat{\boldsymbol{w}}_\varepsilon)_i}{\partial w_j}=\frac{1}{\sqrt{|\boldsymbol{w}|^2+\varepsilon^2}}\left(\delta_{ij}-\frac{w_iw_j}{|\boldsymbol{w}|^2+\varepsilon^2}\right),\qquad i,j\in\{1,2,3\},\tag{5}$$

with exact eigenvalues

$$\lambda_\parallel(\boldsymbol{w})=\frac{\varepsilon^2}{\left(|\boldsymbol{w}|^2+\varepsilon^2\right)^{3/2}}\in\left(0,\frac{1}{\varepsilon}\right],\qquad\lambda_\perp(\boldsymbol{w})=\frac{1}{\left(|\boldsymbol{w}|^2+\varepsilon^2\right)^{1/2}}\in\left(0,\frac{1}{\varepsilon}\right].\tag{6}$$

Consequently, $\|D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon(\boldsymbol{w})\|_{\mathrm{op}}=\lambda_\perp(\boldsymbol{w})\le\frac{1}{\varepsilon}$ for all $\boldsymbol{w}\in\mathbb{R}^3$, and the gated axis map $\beta(\boldsymbol{w}):=\Phi(|\boldsymbol{w}|/\Omega_c)\hat{\boldsymbol{w}}_\varepsilon$ satisfies for all $\boldsymbol{w}_1,\boldsymbol{w}_2\in\mathbb{R}^3$:

$$|\beta(\boldsymbol{w}_1)-\beta(\boldsymbol{w}_2)|\le L_{\beta,\varepsilon}|\boldsymbol{w}_1-\boldsymbol{w}_2|,\qquad L_{\beta,\varepsilon}:=\frac{C_\Phi}{\Omega_c}+\frac{2}{\varepsilon}.\tag{7}$$

Proof. Applying the product and chain rules to $(\hat{\boldsymbol{w}}_\varepsilon)_i=(|\boldsymbol{w}|^2+\varepsilon^2)^{-1/2}w_i$ with $\frac{\partial|\boldsymbol{w}|^2}{\partial w_j}=2w_j$ yields:

$$\frac{\partial(\hat{\boldsymbol{w}}_\varepsilon)_i}{\partial w_j}=(|\boldsymbol{w}|^2+\varepsilon^2)^{-1/2}\delta_{ij}-\frac{1}{2}(|\boldsymbol{w}|^2+\varepsilon^2)^{-3/2}(2w_j)w_i=\frac{1}{\sqrt{|\boldsymbol{w}|^2+\varepsilon^2}}\left(\delta_{ij}-\frac{w_iw_j}{|\boldsymbol{w}|^2+\varepsilon^2}\right),$$

which proves Equation (5). If $\boldsymbol{w}=\boldsymbol{0}$, then $D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon(\boldsymbol{0})=\frac{1}{\varepsilon}I_3$, so all three eigenvalues equal $\frac{1}{\varepsilon}$. If $\boldsymbol{w}\neq\boldsymbol{0}$, let $\boldsymbol{e}_\parallel:=\boldsymbol{w}/|\boldsymbol{w}|$ be the unit vector parallel to $\boldsymbol{w}$, and let $\boldsymbol{e}_{\perp,1},\boldsymbol{e}_{\perp,2}$ be any two orthonormal vectors orthogonal to $\boldsymbol{w}$ ($\boldsymbol{w}\cdot\boldsymbol{e}_{\perp,k}=0$). Then:

$$\begin{aligned}\big(D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon(\boldsymbol{w})\big)\boldsymbol{e}_\parallel&=\frac{1}{\sqrt{|\boldsymbol{w}|^2+\varepsilon^2}}\left(1-\frac{|\boldsymbol{w}|^2}{|\boldsymbol{w}|^2+\varepsilon^2}\right)\boldsymbol{e}_\parallel=\frac{\varepsilon^2}{(|\boldsymbol{w}|^2+\varepsilon^2)^{3/2}}\boldsymbol{e}_\parallel=\lambda_\parallel(\boldsymbol{w})\boldsymbol{e}_\parallel,\\ \big(D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon(\boldsymbol{w})\big)\boldsymbol{e}_{\perp,k}&=\frac{1}{\sqrt{|\boldsymbol{w}|^2+\varepsilon^2}}\boldsymbol{e}_{\perp,k}=\lambda_\perp(\boldsymbol{w})\boldsymbol{e}_{\perp,k},\qquad k\in\{1,2\}.\end{aligned}$$

Since $0<\lambda_\parallel(\boldsymbol{w})\le\lambda_\perp(\boldsymbol{w})\le\frac{1}{\varepsilon}$, the spectral norm of the symmetric matrix $D_{\boldsymbol{w}}\hat{\boldsymbol{w}}_\varepsilon(\boldsymbol{w})$ is bounded by $\frac{1}{\varepsilon}$ (and by the triangle inequality $|\hat{\boldsymbol{w}}_{1,\varepsilon}-\hat{\boldsymbol{w}}_{2,\varepsilon}|\le\frac{2}{\varepsilon}|\boldsymbol{w}_1-\boldsymbol{w}_2|$, as mechanized in `Theorem5_Monotone_Uniqueness.lean`). Combining this with the cutoff Lipschitz estimate

$$\left|\Phi\left(\frac{|\boldsymbol{w}_1|}{\Omega_c}\right)-\Phi\left(\frac{|\boldsymbol{w}_2|}{\Omega_c}\right)\right|\le\frac{C_\Phi}{\Omega_c}||\boldsymbol{w}_1|-|\boldsymbol{w}_2||\le\frac{C_\Phi}{\Omega_c}|\boldsymbol{w}_1-\boldsymbol{w}_2|$$

and $|\hat{\boldsymbol{w}}_{1,\varepsilon}|\le1$, $\Phi\le1$ establishes Inequality (7). □

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3 Convex Potential Structure, Hessian Positivity, and Three-Zone Cubic Coercivity

This section provides the complete analytical proofs for Lemma 3.3 and Lemma 3.4 of the main text, corresponding to `Theorem2_Convex_Potential.lean`, `Theorem3_Cubic_Coercivity.lean`, `CAS_Proof2_Convex_Potential_Hessian.py`, and `CAS_Proof3_Cubic_Threshold_Coercivity.py`.

3.1 Exact Eigenvalue Decomposition of the Potential Hessian $D^2\Psi(\boldsymbol{w}) = DV(\boldsymbol{w})$

Proposition 3.1 (Convexity and Spectral Coercivity of $DV(\boldsymbol{w})$)**.** *Let $h(s) = \nu_{\mathrm{TZT}}(s) = C_O\ell^2 s\Phi(s/\Omega_c)$ for $s \geq 0$, and let $\Psi(\boldsymbol{w}) = \int_0^{|\boldsymbol{w}|} h(r)r\,\mathrm{d}r$ with gradient flux $V(\boldsymbol{w}) = \nabla_{\boldsymbol{w}}\Psi(\boldsymbol{w}) = h(|\boldsymbol{w}|)\boldsymbol{w}$. Then for every $\boldsymbol{w} \in \mathbb{R}^3$, the Hessian matrix $D^2\Psi(\boldsymbol{w}) = DV(\boldsymbol{w})$ has components*

$$(DV(\boldsymbol{w}))_{ij} = \frac{\partial V_i}{\partial w_j}(\boldsymbol{w}) = h(|\boldsymbol{w}|)\delta_{ij} + h'(|\boldsymbol{w}|)\frac{w_i w_j}{|\boldsymbol{w}|}, \qquad i,j \in \{1,2,3\}, \tag{8}$$

with longitudinal eigenvalue $\mu_{\parallel}(\boldsymbol{w})$ (along $\boldsymbol{w}$) and transverse eigenvalues $\mu_{\perp}(\boldsymbol{w})$ (orthogonal to $\boldsymbol{w}$) given by:

$$\mu_{\parallel}(\boldsymbol{w}) = h(|\boldsymbol{w}|) + |\boldsymbol{w}|h'(|\boldsymbol{w}|) = C_O\ell^2|\boldsymbol{w}|\left[2\Phi\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right) + \frac{|\boldsymbol{w}|}{\Omega_c}\Phi'\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right)\right] \geq 0, \tag{9}$$

$$\mu_{\perp}(\boldsymbol{w}) = h(|\boldsymbol{w}|) = C_O\ell^2|\boldsymbol{w}|\Phi\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right) \geq 0. \tag{10}$$

Consequently, for every $\boldsymbol{\xi} \in \mathbb{R}^3$,

$$\boldsymbol{\xi}^{\mathsf{T}} DV(\boldsymbol{w})\boldsymbol{\xi} \geq \mu_{\perp}(\boldsymbol{w})|\boldsymbol{\xi}|^2 = C_O\ell^2\Phi\left(\frac{|\boldsymbol{w}|}{\Omega_c}\right)|\boldsymbol{w}||\boldsymbol{\xi}|^2 \geq 0, \tag{11}$$

and V satisfies the global Minty–Browder monotonicity inequality $(V(\boldsymbol{a}) - V(\boldsymbol{b})) \cdot (\boldsymbol{a} - \boldsymbol{b}) \geq 0$ for all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^3$.

Proof. For $|\boldsymbol{w}| \leq \Omega_c$, $\Phi(|\boldsymbol{w}|/\Omega_c) = 0$ identically, so $\Psi(\boldsymbol{w}) = 0$, $V(\boldsymbol{w}) = \boldsymbol{0}$, and $DV(\boldsymbol{w}) = \boldsymbol{0}$, making Ψ smooth (C^∞) in a neighborhood of the origin $\boldsymbol{w} = \boldsymbol{0}$. For $\boldsymbol{w} \neq \boldsymbol{0}$, differentiating $|\boldsymbol{w}| = \sqrt{w_1^2 + w_2^2 + w_3^2}$ gives $\frac{\partial|\boldsymbol{w}|}{\partial w_j} = \frac{w_j}{|\boldsymbol{w}|}$. Applying the chain rule to $\Psi(\boldsymbol{w}) = \psi(|\boldsymbol{w}|)$ where $\psi'(s) = sh(s)$ gives:

$$\frac{\partial\Psi}{\partial w_i}(\boldsymbol{w}) = \psi'(|\boldsymbol{w}|)\frac{w_i}{|\boldsymbol{w}|} = |\boldsymbol{w}|h(|\boldsymbol{w}|)\frac{w_i}{|\boldsymbol{w}|} = h(|\boldsymbol{w}|)w_i = V_i(\boldsymbol{w}).$$

Differentiating $V_i(\boldsymbol{w}) = h(|\boldsymbol{w}|)w_i$ once more with respect to w_j yields:

$$\frac{\partial V_i}{\partial w_j}(\boldsymbol{w}) = h(|\boldsymbol{w}|)\delta_{ij} + h'(|\boldsymbol{w}|)\frac{w_i w_j}{|\boldsymbol{w}|},$$

where $h'(s) = C_O\ell^2\big[\Phi(s/\Omega_c) + (s/\Omega_c)\Phi'(s/\Omega_c)\big]$. Because $\Phi \geq 0$ and $\Phi' \geq 0$ on $\mathbb{R}$, both $h(s) \geq 0$ and $h'(s) \geq 0$ for all $s \geq 0$.

Let $\boldsymbol{e}_{\parallel} = \boldsymbol{w}/|\boldsymbol{w}|$ and $\boldsymbol{e}_{\perp,1}, \boldsymbol{e}_{\perp,2} \perp \boldsymbol{w}$. Multiplying $DV(\boldsymbol{w})$ by $\boldsymbol{e}_{\parallel}$ gives $(h(|\boldsymbol{w}|) + |\boldsymbol{w}|h'(|\boldsymbol{w}|))\boldsymbol{e}_{\parallel} = \mu_{\parallel}(\boldsymbol{w})\boldsymbol{e}_{\parallel}$, and multiplying by $\boldsymbol{e}_{\perp,k}$ gives $h(|\boldsymbol{w}|)\boldsymbol{e}_{\perp,k} = \mu_{\perp}(\boldsymbol{w})\boldsymbol{e}_{\perp,k}$. Since $h'(|\boldsymbol{w}|) \geq 0$, we have $\mu_{\parallel}(\boldsymbol{w}) \geq \mu_{\perp}(\boldsymbol{w}) = h(|\boldsymbol{w}|) \geq 0$, which proves Inequality (11) (verified symbolically in `CAS_Proof2_Convex_Potential_Hessian.py` and formally in `Theorem2_Convex_Potential.lean`). Finally, for any $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^3$, setting $\boldsymbol{\gamma}(\theta) := \boldsymbol{b} + \theta(\boldsymbol{a} - \boldsymbol{b})$ for $\theta \in [0,1]$ gives:

$$\big(V(\boldsymbol{a}) - V(\boldsymbol{b})\big) \cdot (\boldsymbol{a} - \boldsymbol{b}) = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}\theta}V(\boldsymbol{\gamma}(\theta)) \cdot (\boldsymbol{a} - \boldsymbol{b})\,\mathrm{d}\theta = \int_0^1 (\boldsymbol{a} - \boldsymbol{b})^{\mathsf{T}} DV(\boldsymbol{\gamma}(\theta))(\boldsymbol{a} - \boldsymbol{b})\,\mathrm{d}\theta \geq 0. \quad \square$$

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3.2 Three-Zone Case Analysis of the Pointwise Cubic Coercivity Bound

Proposition 3.2 (Three-Zone Verification of Cubic Coercivity)**.** *For every $\Omega_c > 0$, $C_O > 0$, $\ell > 0$, and every vorticity vector $\boldsymbol{w} \in \mathbb{R}^3$ with magnitude $s = |\boldsymbol{w}| \geq 0$:*

$$s^3 \leq \Phi\left(\frac{s}{\Omega_c}\right) s^3 + 8\Omega_c^3 \qquad \Longleftrightarrow \qquad V(\boldsymbol{w}) \cdot \boldsymbol{w} \geq C_O \ell^2 \left(|\boldsymbol{w}|^3 - 8\Omega_c^3\right). \tag{12}$$

Proof. We verify Inequality (12) across all three regimes of Definition 2.1 (formalized in `Theorem3_Cubic_Coercivity.lean` and `CAS_Proof3_Cubic_Threshold_Coercivity.py`):

1. **Zone I (Subcritical Laminar Zone, $0 \leq s \leq \Omega_c$):** Here $\Phi(s/\Omega_c) = 0$ and $s \leq \Omega_c < 2\Omega_c$, so $s^3 \leq \Omega_c^3 \leq 8\Omega_c^3 = \Phi(s/\Omega_c)s^3 + 8\Omega_c^3$.
2. **Zone II (Transition Shell, $\Omega_c < s < 2\Omega_c$):** Here $\Phi(s/\Omega_c) \in [0,1]$ and $s < 2\Omega_c$, so $\big(1 - \Phi(s/\Omega_c)\big)s^3 \leq (2\Omega_c)^3 = 8\Omega_c^3$, which rearranges to $s^3 \leq \Phi(s/\Omega_c)s^3 + 8\Omega_c^3$.
3. **Zone III (Supercritical Active Core, $s \geq 2\Omega_c$):** Here $\Phi(s/\Omega_c) = 1$, so $\big(1 - \Phi(s/\Omega_c)\big)s^3 = 0 \leq 8\Omega_c^3$, and thus $s^3 = \Phi(s/\Omega_c)s^3 \leq \Phi(s/\Omega_c)s^3 + 8\Omega_c^3$.

Multiplying by $C_O\ell^2 > 0$ and noting $V(\boldsymbol{w}) \cdot \boldsymbol{w} = C_O\ell^2\Phi(|\boldsymbol{w}|/\Omega_c)|\boldsymbol{w}|^3$ completes the proof. □

4 Componentwise Derivation and Energetics of the Rotational-to-Strain Subgrid Identity

This section provides the complete tensor-index proof and energy-level comparison for Equation (28) and Appendix A (Lemma A.1) of the main text, corresponding to `Theorem4_Rotational_Identity.lean` and `CAS_Proof4_Rotational_Strain_Identity.py`.

Theorem 4.1 (Exact Rotational-to-Strain Vector Calculus Identity and Global Dissipation Equivalence)**.** *Let $\boldsymbol{u} \in H^2_\sigma(\mathbb{T}^3)$ be any divergence-free velocity field ($\partial_j u_j = 0$) with vorticity $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ ($\omega_k = \varepsilon_{klm}\partial_l u_m$) and symmetric strain tensor $S_{ij}(\boldsymbol{u}) = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$. For any scalar viscosity field $\nu_t \in C^1(\mathbb{T}^3)$:*

1. ***Pointwise Vector Identity:*** *For each Cartesian component $i \in \{1,2,3\}$,*

$$-\big(\nabla \times (\nu_t \boldsymbol{\omega})\big)_i = \partial_j\big(2\nu_t S_{ij}(\boldsymbol{u})\big) - 2(\partial_i u_j)(\partial_j \nu_t). \tag{13}$$

2. ***Global Energy Dissipation Identity:*** *When tested against $\boldsymbol{u}$ in $L^2(\mathbb{T}^3)$, the rotational dissipation operator $F_{\text{rot}} := -\nabla \times (\nu_t \boldsymbol{\omega})$ and the classical symmetric strain operator $F_{\text{sym}} := \nabla \cdot (2\nu_t S(\boldsymbol{u}))$ satisfy:*

$$\begin{aligned} \langle -\nabla \times (\nu_t \boldsymbol{\omega}), \boldsymbol{u} \rangle_{L^2(\mathbb{T}^3)} &= -\int_{\mathbb{T}^3} \nu_t(x) |\boldsymbol{\omega}(x)|^2 \, \mathrm{d}x, \\ \langle \nabla \cdot (2\nu_t S(\boldsymbol{u})), \boldsymbol{u} \rangle_{L^2(\mathbb{T}^3)} &= -\int_{\mathbb{T}^3} 2\nu_t(x) |S(\boldsymbol{u})(x)|^2 \, \mathrm{d}x, \end{aligned} \tag{14}$$

whose difference equals the boundary-free commutator integral $\int_{\mathbb{T}^3} 2u_i(\partial_i u_j)(\partial_j \nu_t) \, \mathrm{d}x$, which vanishes identically whenever ν_t is spatially constant ($\nabla \nu_t = \mathbf{0}$).

Proof. (1) Using the summation convention over repeated indices $j,k,l,m \in \{1,2,3\}$ and the Levi-Civita contraction identity $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$, we expand the i-th component of $-\nabla \times (\nu_t \boldsymbol{\omega})$:

$$\begin{aligned} -\big(\nabla \times (\nu_t \boldsymbol{\omega})\big)_i &= -\varepsilon_{ijk}\partial_j(\nu_t \omega_k) = -\varepsilon_{ijk}\varepsilon_{klm}\partial_j\big(\nu_t \partial_l u_m\big) \\ &= -\big(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}\big)\big((\partial_j \nu_t)(\partial_l u_m) + \nu_t \partial_j \partial_l u_m\big) \\ &= -\big[(\partial_j \nu_t)(\partial_i u_j) + \nu_t \partial_j \partial_i u_j\big] + \big[(\partial_j \nu_t)(\partial_j u_i) + \nu_t \partial_j \partial_j u_i\big] \\ &= \nu_t \Delta u_i - \nu_t \partial_i(\partial_j u_j) + (\partial_j \nu_t)(\partial_j u_i) - (\partial_j \nu_t)(\partial_i u_j). \end{aligned}$$

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On the other hand, expanding the divergence of the symmetric strain flux $\partial_j(2\nu_t S_{ij}(\boldsymbol{u}))$ yields:

$$\begin{aligned}\partial_j\big(2\nu_t S_{ij}(\boldsymbol{u})\big) &= \partial_j\big[\nu_t(\partial_i u_j + \partial_j u_i)\big] \\ &= (\partial_j\nu_t)(\partial_i u_j) + (\partial_j\nu_t)(\partial_j u_i) + \nu_t\partial_j\partial_i u_j + \nu_t\partial_j\partial_j u_i \\ &= \nu_t\Delta u_i + \nu_t\partial_i(\partial_j u_j) + (\partial_j\nu_t)(\partial_j u_i) + (\partial_j\nu_t)(\partial_i u_j).\end{aligned}$$

Subtracting the second expansion from the first gives:

$$-\big(\nabla\times(\nu_t\boldsymbol{\omega})\big)_i - \partial_j\big(2\nu_t S_{ij}(\boldsymbol{u})\big) = -2\nu_t\partial_i(\nabla\cdot\boldsymbol{u}) - 2(\partial_i u_j)(\partial_j\nu_t).$$

Since $\nabla\cdot\boldsymbol{u} = \partial_j u_j = 0$, the divergence gradient term $-2\nu_t\partial_i(\nabla\cdot\boldsymbol{u})$ vanishes identically, establishing Equation (13) (verified component-by-component with zero symbolic residual in `CAS_Proof4_Rotational_Strain_Identity.py`).

(2) Integrating by parts on the periodic boundaryless torus $\mathbb{T}^3$ gives

$$\langle -\nabla\times(\nu_t\boldsymbol{\omega}), \boldsymbol{u}\rangle = -\int_{\mathbb{T}^3}(\nu_t\boldsymbol{\omega})\cdot(\nabla\times\boldsymbol{u})\,\mathrm{d}x = -\int_{\mathbb{T}^3}\nu_t|\boldsymbol{\omega}|^2\,\mathrm{d}x,$$

and using the symmetry $S_{ij} = S_{ji}$ yields

$$\langle\nabla\cdot(2\nu_t S(\boldsymbol{u})), \boldsymbol{u}\rangle = -\int_{\mathbb{T}^3}2\nu_t S_{ij}(\partial_j u_i)\,\mathrm{d}x = -\int_{\mathbb{T}^3}2\nu_t S_{ij}S_{ij}\,\mathrm{d}x = -\int_{\mathbb{T}^3}2\nu_t|S(\boldsymbol{u})|^2\,\mathrm{d}x.$$

When ν_t is constant, Lemma 1.1 ($\|\boldsymbol{\omega}\|_{L^2}^2 = 2\|S(\boldsymbol{u})\|_{L^2}^2$) confirms that the two integrals coincide. □

5 Expanded Proofs of Global Strong Well-Posedness, Uniqueness, and Vanishing Filter Limit

In this section, we provide the expanded functional-analytic derivations supporting Theorems 4.1, 4.2, and 4.3 of the main manuscript, corresponding to `Theorem5_Monotone_Uniqueness.lean` and `Theorem6_GronwallFree_Enstrophy.lean`.

5.1 Detailed Derivation of the Grönwall-Free Enstrophy Inequality and Time-Derivative Regularity

Theorem 5.1 (Expanded Proof of Theorem 4.1: Grönwall-Free Enstrophy Closure and $L^{3/2}$ Time-Derivative Bound)**.** *Under the hypotheses of Theorem 4.1 of the main text, let $\boldsymbol{u}_N \in C^1([0,\infty);\mathcal{V}_N)$ be the N-th spectral Galerkin solution to Equation (18) of the main text with initial data $\boldsymbol{u}_N(0) = \mathbb{P}_N\boldsymbol{u}_0 \in H^1_\sigma(\mathbb{T}^3)$ and vorticity $\boldsymbol{\omega}_N = \nabla\times\boldsymbol{u}_N$. Then for every $T > 0$:*

1. *The sequence $\{\boldsymbol{u}_N\}$ satisfies the uniform Grönwall-free enstrophy bound:*

$$\begin{aligned}\sup_{t\in[0,T]}\|\boldsymbol{\omega}_N(t)\|_{L^2}^2 + \nu\int_0^T\|\nabla\boldsymbol{\omega}_N(s)\|_{L^2}^2\,\mathrm{d}s + 2C_O\ell^2\int_0^T\int_{\mathbb{T}^3}\Phi\left(\frac{|\boldsymbol{\omega}_N|}{\Omega_c}\right)|\boldsymbol{\omega}_N||\nabla\boldsymbol{\omega}_N|^2\,\mathrm{d}x\,\mathrm{d}s \\ \leq \|\boldsymbol{\omega}_0\|_{L^2}^2 + 2C_3\left(\frac{\mathcal{E}(T)}{C_O\ell^2} + 8\Omega_c^3(2\pi)^3T\right) \\ + \frac{1}{\nu}\int_0^T\left(|c_X|\Omega_c M(s) + \|\boldsymbol{f}(s)\|_{L^2}\right)^2\mathrm{d}s =: \mathcal{C}_1(T) < \infty.\end{aligned} \tag{15}$$

2. *The sequence of rotational dissipation terms $\{\nabla\times V(\boldsymbol{\omega}_N)\}$ and time derivatives $\{\partial_t\boldsymbol{u}_N\}$ is uniformly bounded in $L^{3/2}((0,T)\times\mathbb{T}^3)$:*

$$\|\nabla\times V(\boldsymbol{\omega}_N)\|_{L^{3/2}((0,T)\times\mathbb{T}^3)} + \|\partial_t\boldsymbol{u}_N\|_{L^{3/2}((0,T)\times\mathbb{T}^3)} \leq \mathcal{C}_2(T) < \infty, \tag{16}$$

with $\mathcal{C}_1(T), \mathcal{C}_2(T)$ completely independent of the Galerkin truncation dimension N.

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Proof. (1) From Step 1 and Step 2 of Theorem 4.1 in the main text, testing the Galerkin ODE with $\boldsymbol{u}_N$ yields $\|\boldsymbol{u}_N(t)\|_{L^2} \le M(t)$ and the uniform space-time cubic vorticity bound:

$$\int_0^T \|\boldsymbol{\omega}_N(s)\|_{L^3(\mathbb{T}^3)}^3 \,\mathrm{d}s \le \frac{\mathcal{E}(T)}{C_O \ell^2} + 8\Omega_c^3 (2\pi)^3 T.$$

Testing the Galerkin ODE with $-\Delta \boldsymbol{u}_N = \nabla \times \boldsymbol{\omega}_N \in \mathcal{V}_N$, we examine the convective inner product $\langle -(\boldsymbol{u}_N \cdot \nabla)\boldsymbol{u}_N, \nabla \times \boldsymbol{\omega}_N \rangle$. Using the vector calculus identity $(\boldsymbol{u}_N \cdot \nabla)\boldsymbol{u}_N = \nabla\left(\frac{1}{2}|\boldsymbol{u}_N|^2\right) + \boldsymbol{\omega}_N \times \boldsymbol{u}_N$ and integrating the curl by parts on $\mathbb{T}^3$ gives:

$$\begin{aligned}\langle -(\boldsymbol{u}_N \cdot \nabla)\boldsymbol{u}_N, \nabla \times \boldsymbol{\omega}_N \rangle &= -\int_{\mathbb{T}^3} \nabla \times \left(\boldsymbol{\omega}_N \times \boldsymbol{u}_N\right) \cdot \boldsymbol{\omega}_N \,\mathrm{d}x \\ &= \int_{\mathbb{T}^3} \left((\boldsymbol{\omega}_N \cdot \nabla)\boldsymbol{u}_N - (\boldsymbol{u}_N \cdot \nabla)\boldsymbol{\omega}_N\right) \cdot \boldsymbol{\omega}_N \,\mathrm{d}x,\end{aligned}$$

where we used $\nabla \cdot \boldsymbol{u}_N = 0$ and $\nabla \cdot \boldsymbol{\omega}_N = 0$. Since $\int_{\mathbb{T}^3} (\boldsymbol{u}_N \cdot \nabla)\boldsymbol{\omega}_N \cdot \boldsymbol{\omega}_N \,\mathrm{d}x = \frac{1}{2}\int_{\mathbb{T}^3} (\boldsymbol{u}_N \cdot \nabla)|\boldsymbol{\omega}_N|^2 \,\mathrm{d}x = 0$ and $(\boldsymbol{\omega}_N \cdot \nabla)\boldsymbol{u}_N \cdot \boldsymbol{\omega}_N = \boldsymbol{\omega}_N \cdot S(\boldsymbol{u}_N)\boldsymbol{\omega}_N$, Lemma 3.5 (the periodic Calderón–Zygmund estimate at $p = 3$) bounds this term by $C_3 \|\boldsymbol{\omega}_N\|_{L^3}^3$. Combining this with Young's inequality on the linear terms and integrating over $[0, T]$ directly yields Estimate (15) (the algebraic skeleton of which is formally verified in `Theorem6_GronwallFree_Enstrophy.lean`).

(2) To establish Estimate (16), observe first that by Sobolev embedding $H^1(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$ in 3D, Estimate (15) implies that $\{\boldsymbol{\omega}_N\}$ is uniformly bounded in $L^\infty(0,T; L^2(\mathbb{T}^3)) \cap L^2(0,T; L^6(\mathbb{T}^3))$. By standard Gagliardo–Nirenberg interpolation:

$$\begin{aligned}&\|\boldsymbol{\omega}_N\|_{L^{10/3}(\mathbb{T}^3)} \le \|\boldsymbol{\omega}_N\|_{L^2(\mathbb{T}^3)}^{2/5} \|\boldsymbol{\omega}_N\|_{L^6(\mathbb{T}^3)}^{3/5} \\ \Longrightarrow \quad &\int_0^T \|\boldsymbol{\omega}_N(s)\|_{L^{10/3}(\mathbb{T}^3)}^{10/3} \,\mathrm{d}s \le \left(\sup_{s\in[0,T]} \|\boldsymbol{\omega}_N(s)\|_{L^2}^2\right)^{2/3} \int_0^T \|\boldsymbol{\omega}_N(s)\|_{L^6(\mathbb{T}^3)}^2 \,\mathrm{d}s \le C\mathcal{C}_1(T)^{5/3}.\end{aligned}$$

In particular, since $\frac{10}{3} > 3$, $\{\boldsymbol{\omega}_N\}$ is uniformly bounded in $L^{10/3}((0,T)\times\mathbb{T}^3)$ and in $L^3((0,T)\times\mathbb{T}^3)$.

Next, we compute the spatial gradient of $V(\boldsymbol{\omega}_N) = h(|\boldsymbol{\omega}_N|)\boldsymbol{\omega}_N$. Since $V(\boldsymbol{\omega}_N) = \mathbf{0}$ whenever $|\boldsymbol{\omega}_N| \le \Omega_c$, $\nabla V(\boldsymbol{\omega}_N)$ is supported on the set $\{|\boldsymbol{\omega}_N| \ge \Omega_c\}$. Using $|h(s)| + s|h'(s)| \le C_O\ell^2(2 + 2C_\Phi)s$ and dividing into the transition zone $\Omega_c \le |\boldsymbol{\omega}_N| \le 2\Omega_c$ and the core zone $|\boldsymbol{\omega}_N| \ge 2\Omega_c$ (where $\Phi(|\boldsymbol{\omega}_N|/\Omega_c) = 1$), we obtain pointwise:

$$|\nabla \times V(\boldsymbol{\omega}_N)| \le \sqrt{3}|\nabla V(\boldsymbol{\omega}_N)| \le C\ell^2 \left(\Omega_c |\nabla \boldsymbol{\omega}_N| + |\boldsymbol{\omega}_N|^{1/2} \left(\Phi\left(\frac{|\boldsymbol{\omega}_N|}{\Omega_c}\right)^{1/2} |\boldsymbol{\omega}_N|^{1/2} |\nabla \boldsymbol{\omega}_N|\right)\right).$$

On $(0,T)\times\mathbb{T}^3$, $\Omega_c \nabla \boldsymbol{\omega}_N$ is uniformly bounded in $L^2((0,T)\times\mathbb{T}^3) \hookrightarrow L^{3/2}((0,T)\times\mathbb{T}^3)$. For the second term, let $A_N := |\boldsymbol{\omega}_N|^{1/2}$ and $B_N := \Phi(|\boldsymbol{\omega}_N|/\Omega_c)^{1/2}|\boldsymbol{\omega}_N|^{1/2}|\nabla\boldsymbol{\omega}_N|$. From the third term on the left-hand side of Estimate (15), $\|B_N\|_{L^2((0,T)\times\mathbb{T}^3)}^2 \le (2C_O\ell^2)^{-1}\mathcal{C}_1(T) < \infty$, while $\|A_N\|_{L^6((0,T)\times\mathbb{T}^3)}^6 = \int_0^T \int_{\mathbb{T}^3} |\boldsymbol{\omega}_N|^3 \,\mathrm{d}x\,\mathrm{d}s < \infty$. Applying Hölder's inequality with exponents $\frac{1}{2/3} = \frac{1}{6} + \frac{1}{2}$ (i.e., $p = 4$ and $p' = \frac{4}{3}$ on $|A_N|^{3/2}|B_N|^{3/2}$) gives:

$$\int_0^T \int_{\mathbb{T}^3} |A_N B_N|^{3/2} \,\mathrm{d}x\,\mathrm{d}s \le \left(\int_0^T \int_{\mathbb{T}^3} |A_N|^6 \,\mathrm{d}x\,\mathrm{d}s\right)^{1/4} \left(\int_0^T \int_{\mathbb{T}^3} |B_N|^2 \,\mathrm{d}x\,\mathrm{d}s\right)^{3/4} < \infty.$$

Since $\mathbb{P}_N$ is uniformly bounded on $L^{3/2}(\mathbb{T}^3)$ and all remaining terms in the Galerkin ODE ($\nu\Delta\boldsymbol{u}_N$, $\mathbb{P}_N[(\boldsymbol{u}_N \cdot \nabla)\boldsymbol{u}_N]$, $c_X\Omega_c\mathbb{P}_N G(\boldsymbol{u}_N, \boldsymbol{\omega}_N)$, $\mathbb{P}_N \boldsymbol{f}$) are uniformly bounded in $L^2(0,T; L^2(\mathbb{T}^3)) \hookrightarrow L^{3/2}((0,T)\times\mathbb{T}^3)$, Estimate (16) follows immediately. □

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6 Verification Mapping to Lean 4 (Supplementary_2) and SymPy CAS (Supplementary_3)

Table 1 provides the complete cross-reference between the analytical propositions proved in this document, the theorems of the main manuscript (Part3_Paper_v1.5-100.tex), the mechanized Lean 4 formal verification files (Supplementary_2_Formal_Verification_Lean4), and the executable SymPy CAS scripts (Supplementary_3_CAS_Python_Proofs).

Table 1: Complete cross-reference matrix linking analytical derivations in Supplementary_1, main text theorems, Lean 4 formal proofs (Supplementary_2), and SymPy CAS symbolic scripts (Supplementary_3).

Supplementary 1	**Main Text Ref.**	**Lean 4 File (Supplementary_2)**	**SymPy CAS File (Supplementary_3)**
Lemmas 1.1, 1.2, 1.3	Defs. 2.1–2.4, Eq. (16)	Theorem0_Basic_Operators.lean	—
Proposition 2.1	Lemma 3.1	Theorem1_Gyroscopic_Neutrality.lean	CAS_Proof1_Gyroscopic_Neutrality.py
Proposition 2.2	Lemma 3.2, Theorem 4.2	Theorem5_Monotone_Uniqueness.lean	CAS_Proof1_Gyroscopic_Neutrality.py
Proposition 3.1	Lemma 3.3(1)–(2), Lemma 3.4	Theorem2_Convex_Potential.lean	CAS_Proof2_Convex_Potential_Hessian.py
Proposition 3.2	Lemma 3.3(3), Eq. (20)	Theorem3_Cubic_Coercivity.lean	CAS_Proof3_Cubic_Threshold_Coercivity.py
Theorem 4.1	Eq. (28), Lemma A.1	Theorem4_Rotational_Identity.lean	CAS_Proof4_Rotational_Strain_Identity.py
Theorem 5.1	Theorems 4.1–4.3, Lemma A.2	Theorem6_GronwallFree_Enstrophy.lean	CAS_Proof3_Cubic_Threshold_Coercivity.py

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Declarations

Data and Code Availability. All mechanized Lean 4 formal verification suites, symbolic CAS scripts, numerical solvers, and benchmark datasets supporting the findings of this study are publicly archived via the canonical repository (`https://github.com/arkhenomia/arkheion`) and the dataset archive (`https://huggingface.co/datasets/arkhenomia/arkheion`).

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